Inventory of employer and union relationships in the Collective Agreements e-Library Portal

1.Introduction

1.1 Purpose of the Study

The Collective Agreements e-Library Portal contains detailed information on employer–union relationships across multiple industrial sectors. This project analyzes the dataset to identify underlying patterns in collective bargaining agreements using distributed data processing and machine learning techniques.

Apache Spark was used to process and analyze the data due to its scalability and support for machine learning through Spark MLlib. K-Means clustering was applied to group agreements based on organizational size, contract duration, and sector classification.

1.2 Motivation

Understanding patterns in collective agreements can support:

- Policy analysis
- Labor market research
- Strategic workforce planning

By identifying clusters of similar agreements, this project aims to provide insights into how agreement characteristics vary across sectors and employer sizes.

1.3 Data Source

The dataset used in this project is from the Ontario official site. It contains 70,000+ records and 15 columns including 'Employer', 'Union', 'Sector'.

URL:

https://data.ontario.ca/dataset/inventory-of-employer-and-union-relationships-in-the-collective-agreements-e-library-portal

The dataset has 20777 unique employers, 9443 unique unions, 4 different industrial sector types, and 128 different sectors, 358 industries.

2. Objectives

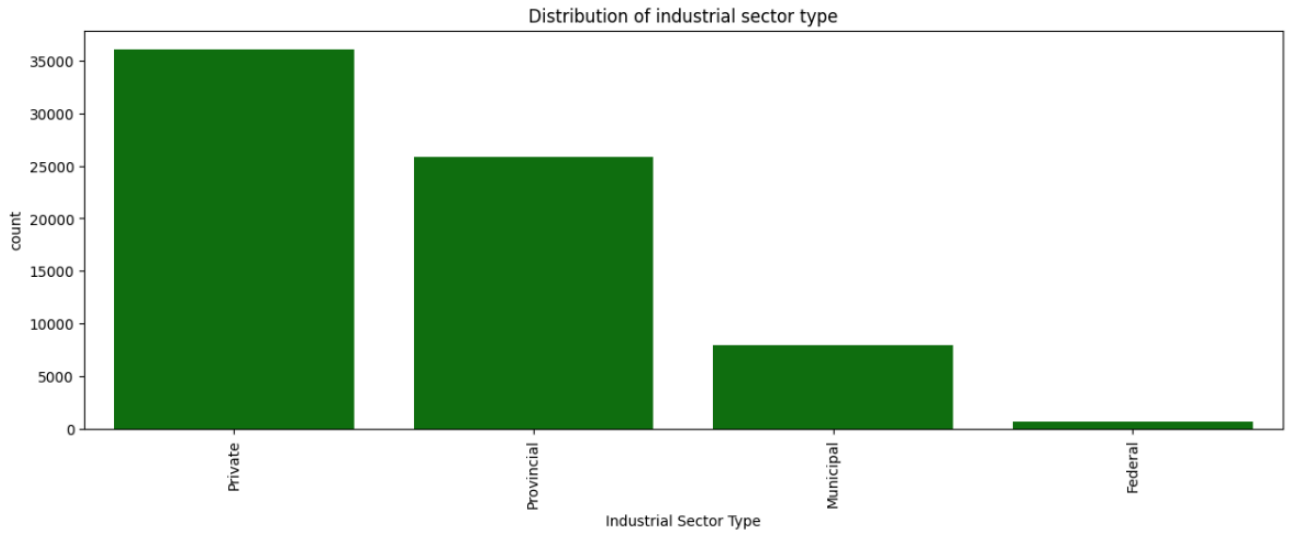
The main objectives of this project are:

2.1 Data Preparation

- Trimmed the Employer values
- Formatted number of employees column values
- Handled missing and inconsistent values

2.2 Exploratory Data Analysis

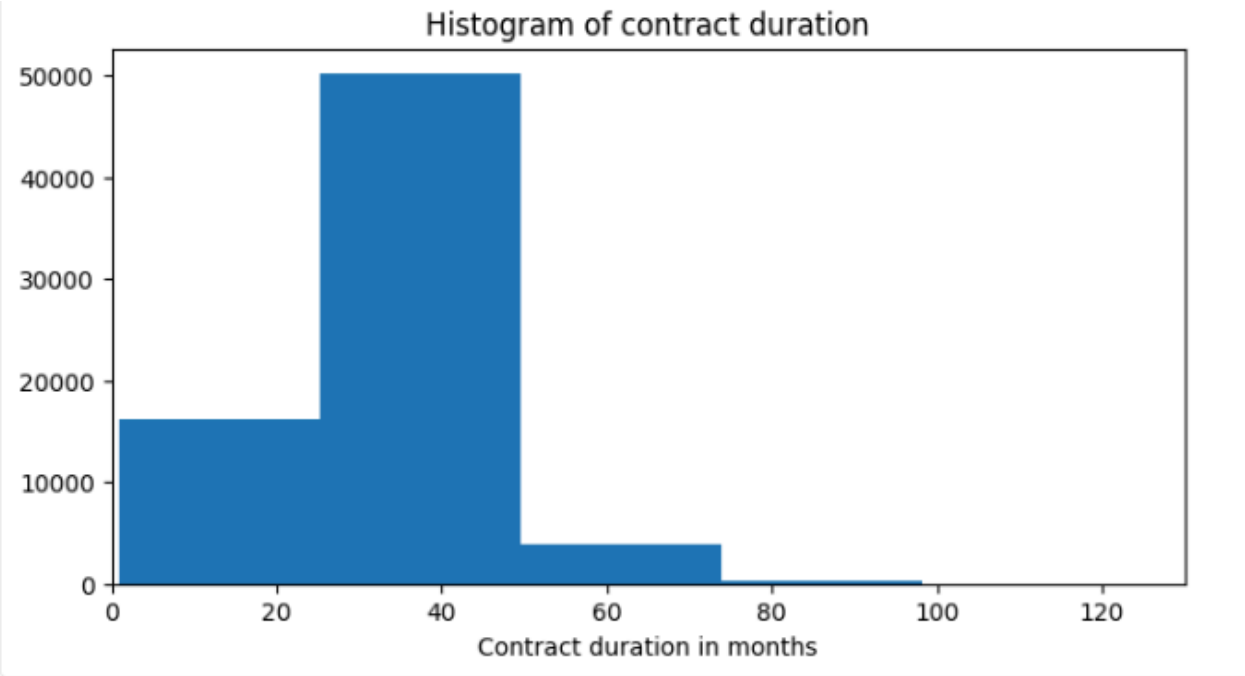
2.2.1 Analyze the distribution of agreements by sector



Insights:

- The Private sector has the highest number of agreements, indicates that collective bargaining is most prevalent in private organizations.
- Provincial agreements are significantly fewer than private, but still substantial, reflects centralized bargaining in education, healthcare, and public services
- Municipal agreements are relatively limited, suggesting localized and standardized bargaining structures.
- Federal-level collective agreements are rare compared to other sectors.

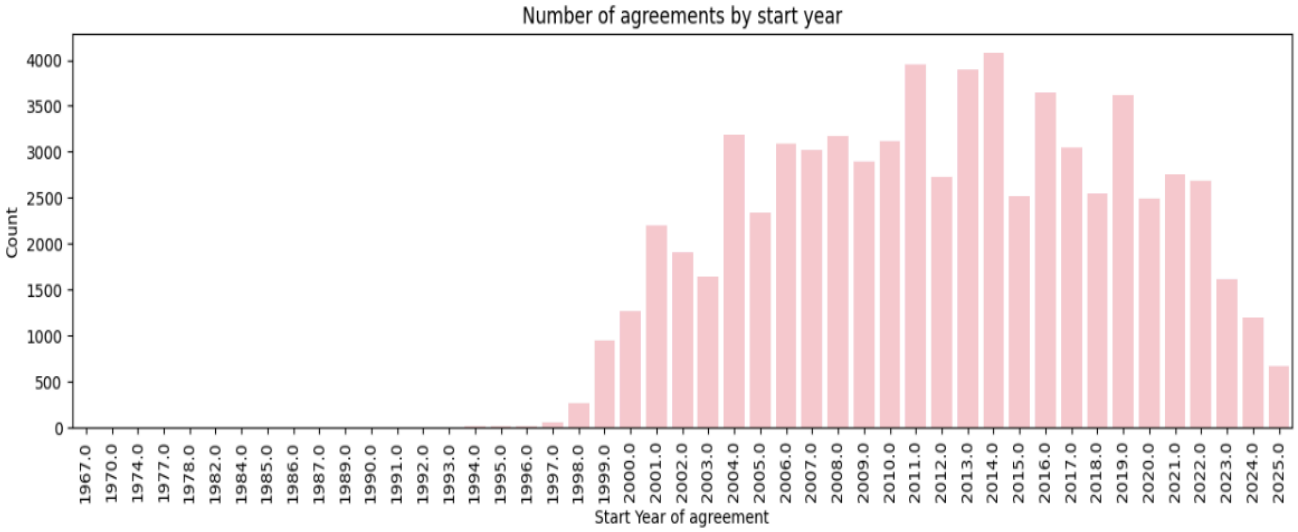
2.2.2 Examine contract duration patterns



Insights:

- Around 50,000 Agreements have 25-45 months of contracts.
- And 15000+ agreements are 0-25 months of contracts in length.

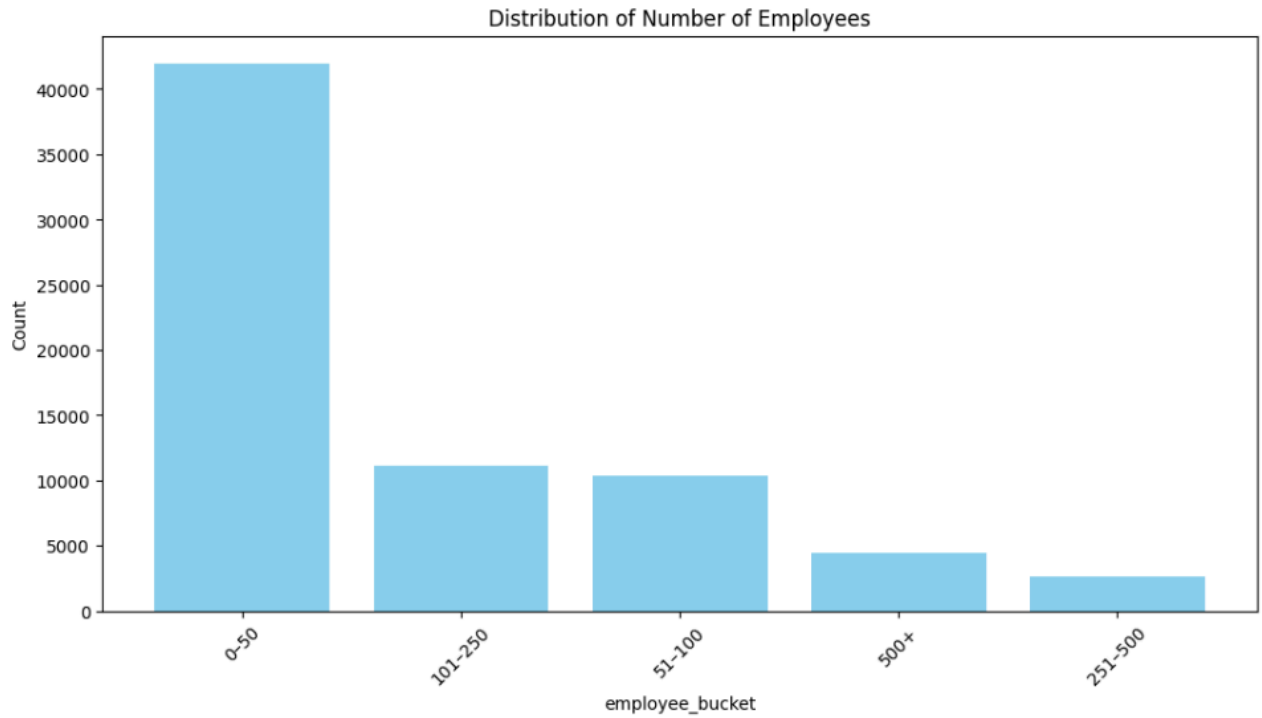
2.2.3 Number of agreements by year



Insights:

- Most of the agreements started between 2001 and 2022.
- Highest agreements started in 2014, around 4000.

2.2.4 Distribution of Number of employees



Interpretation

- Raw numeric values in Number of Employees→ noisy, hard to interpret
- Bucketed values → clearer distribution pattern
- Most agreements involve small organizations (0–50 employees)

2.3 Feature Engineering

- Engineered contract length variable using start and end of agreement date.
- Extracted start year from agreement date.

2.4 Model Development

- **Feature selection for the model**

  - Employee count
  - Contract length
  - Industry sector type

- **Encoding the categorical variable using StringIndexer**

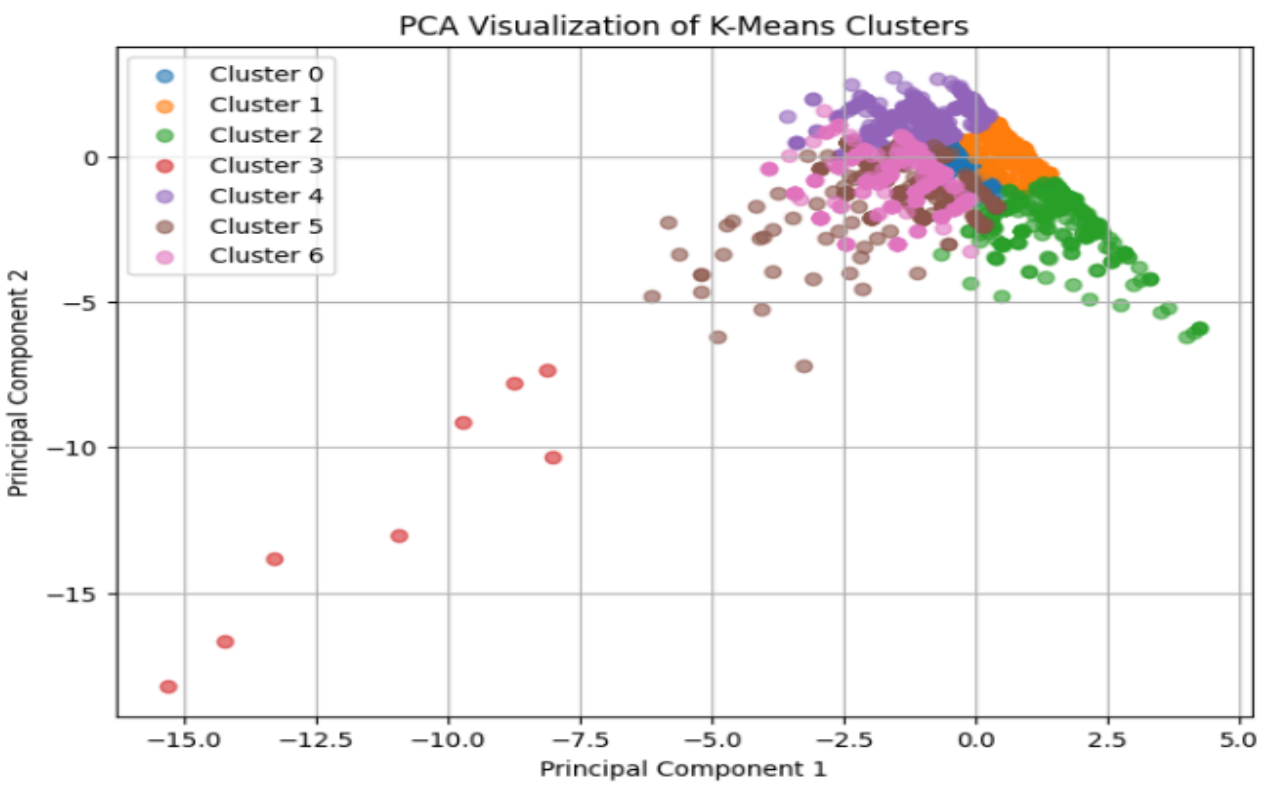
  - Industry sector type

- **Determine the optimal number of clusters using silhouette analysis**

The optimal number of clusters was determined as 8 using the Silhouette score. PCA with two components was applied after clustering solely for visualization purposes and does not affect the clustering outcome.

- **Applied PCA for visualization**

PCA was applied to reduce high-dimensional features into two principal components for visualization. Each point represents a collective agreement, and colors indicate cluster membership identified by the K-Means model. Trained a final clustering model on the full dataset.



Interpretation

K-Means clustering with k=8 revealed distinct groupings of collective agreements. PCA visualization confirms reasonable cluster separation, with one cluster clearly isolated, indicating agreements with unique characteristics.

Clear cluster structure

- Clusters 6 and 5 are compact and well separated
- Indicates that K-Means found meaningful groupings and features used for clustering are informative

Overlapping clusters

- Clusters 1,2, 0 and 4 overlap around `(PC1 ≈ -1 to 1)`
- This is normal and expected because:
    - PCA compresses high-dimensional data into 2D
    - Overlap does not mean poor clustering

"Some overlap is observed due to dimensionality reduction and similarity between agreements."

Cluster 3 (red points)

- Cluster far left & bottom (large negative PC1 & PC2)
- This suggests that outlier group
- Possibly agreements with:
    - Very high or very low employee counts
    - Unusual contract lengths

3. Findings

Dataset Insights

- The majority of collective agreements belong to the private sector, followed by provincial, municipal, and federal sectors.
- Most agreements involve small to medium-sized employers, while large employers are less frequent.

- Contract durations vary widely, with certain sectors showing longer-term agreements.

Clustering Results

- K-Means clustering identified distinct groups of collective agreements based on:
    - Number of employees
    - Contract length
    - Industrial sector classification
- Some clusters were dominated by private-sector agreements with shorter contract durations.
- Other clusters represented public-sector agreements with larger workforces and longer contracts.

Model Evaluation

- Silhouette analysis was used to select the optimal number of clusters.
- The final model demonstrated reasonable cluster separation, indicating meaningful grouping of agreement characteristics.
- Feature scaling significantly improved clustering stability and interpretability.

Limitations

- High-cardinality attributes such as agreement location were excluded from modeling.
- Name attribute contains file name/number and it's excluded from analysis and modeling.

Appendices

Data Schema

```
root
 |-- _id: long (nullable = true)
 |-- Name: string (nullable = true)
 |-- Employer: string (nullable = true)
 |-- Union: string (nullable = true)
 |-- Agreement Start Date: date (nullable = true)
 |-- Agreement Expiry Date: date (nullable = true)
 |-- Industrial Sector Type: string (nullable = true)
 |-- Labour Legislation (Act): string (nullable = true)
 |-- Number of Employees: string (nullable = true)
 |-- Legacy Relationship ID: string (nullable = true)
 |-- Industry: string (nullable = true)
 |-- Sector: string (nullable = true)
 |-- Agreement Location: string (nullable = true)
 |-- OLRB Construction Industry Areas: string (nullable = true)
 |-- Relationship ID: string (nullable = true)
```

Number of K and Silhouette score Table

	k	silhouette
1	2	0.436343871593454...
2	3	0.5401819039885805
3	4	0.5115245299486525
4	5	0.383256146177258...
5	6	0.548313136986909
6	7	0.6109343871198131
7	8	0.6359132851609158